

The PLL circuitry of the chip requires a clean supply and thus it is necessary to isolate the PLL supply from the digital supply, especially when the PLL and digital supplies are derived from a single supply source. I/O switching noise gets introduced in the digital supply and this high switching noise should be avoided from entering into the PLL supply by using appropriate filters such as Hi-Z ferrite bead in series and capacitors in shunt. Otherwise, this high-frequency noise could affect the functionality of the PLL and hence the stability of the clocks generated by the PLL.

Isolation between analogue and digital power and ground planes. Systems having mixed-signal circuitry should ensure effective isolation of power and ground planes between analogue and digital supplies. When an analogue-to-digital converter (ADC) converts a continuous time signal into discrete time digital samples, it basically samples the input signal with reference to ground. High-frequency digital switching noise could get injected into the ground plane, giving rise to fluctuations on the ground plane. Now that the reference keeps fluctuating, the potential difference between the signal and reference (system ground), which is actually quantised, will not be faithful.

The digital ground/power plane and analogue ground/power plane should be isolated by using ferrite beads between digital and analogue planes. Thus under DC condition, digital ground is shorted to analogue ground but the ferrite bead (inductive series filter) avoids high-frequency switching noise (AC component) from getting injected into the analogue ground. The same holds for the power planes.

Used gated clocks for such ADCs so that the digital bit-shift clock is shut off during the conversion time, preventing the noise from getting introduced into the analogue circuitry within or outside the converter IC. The bit clock is active only when the data is transferred from the converter to the

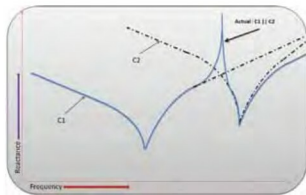


Fig. 4: In reality the circuit behaves as shown here

processor over a serial data link like the SPI.

Critical length, reflections and terminations. This is the most important factor to be considered when routing high-speed signals such as SDRAM and DDR-II. The trace length should not exceed the critical length. This is related to the propagation time of the electrical signal in a particular medium. The electrical signal travels at the speed of light through a wire surrounded by vacuum. But if this wire is surrounded by some other medium, the propagation time would depend on the relative permittivity of the medium. Relative permittivity 'Er' of a commonly used PCB material FR4 is 4.

The propagation speed of an electrical signal in a medium is given by the relationship:

$$\text{Speed} = 11.8 / \sqrt{\epsilon_r} \text{ inch/ns}$$

For $\epsilon_r = 4$, $\sqrt{\epsilon_r} = 2$ and hence the propagation speed is approximately 15.2 cm/ns (6 inches/ns). That means the signal travels 15.2 cm in one nanosecond. Hence when the reflected signal is considered, the round trip time of 1 ns is equal to 7.6cm (3-inch) trace length. For a signal that has a rise time of more than 1 ns, the reflected signal coincides with the forward signal. When reflections coincide with the forward signal, there is not much of a problem. When this happens, the reflected signal sort of gets absorbed in the forward signal. The boundary here is called the critical length, which is equal to 7.6 cm for a signal of 1ns rise time in FR4.

As long as the trace length is less than the critical length, the reflections coincide with the forward signal and this does not cause any trouble. But when the trace length is more than the critical length, the reflected signal aligns with the steady-state (stabilised) signal at the driver end, giving

rise to ringing—which is most likely to cause problem.

To avoid this problem, the designer has to either slow down his signal (increase the rise time) or shorten the trace length. Series resistor termination exactly does the former. Adding a series termination at the source side increases the rise time of the signal (slower rise). This allows the reflected signal to coincide with the rising signal and get absorbed. One can think of it as the inertia of the driver overpowering the reflection. Higher values of the series termination resistor will increase the rise time even further, but by doing so the signal timings become sluggish. This may give rise to timing specification violation such as set-up and hold with reference to other signals.

So technically a better approach would be to maintain the trace lengths shorter (how much shorter is determined by the critical length), depending on the PCB material (hence propagation time of the electrical signal in that material) and the rise time of the signal.

Other considerations. The following design considerations are equally important and must be kept in mind during system design: I/O leakage current specification, glue logic considerations, multifunctional signal pins, power supply, reset and clock input, unused signals, debug port for prototypes and shared resource system.

I/O leakage current specification. It is usually of the order of a few micro-amperes. Under certain interface con-